

Problem Sheet: Chain Rule and Substitutions 2

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Exercise 1. Differentiate the following functions.

1. $\sin(\cos(3x))$
2. $e^{2\sin(4x)}$
3. $\cos(\ln(5x))$
4. $(\ln(2x))^3$

Exercise 2. Let $f(x) = e^{3\sin x} + \sin(x^2)$. Find $f'(x)$ and $f''(x)$.

Exercise 3. For each integral below, state a suitable substitution before evaluating the indefinite integral.

1. $\int x e^{x^2} dx$
2. $\int \frac{4x}{1+2x^2} dx$
3. $\int 6x \sqrt{1-3x^2} dx$